

THE LIVING MAP: SPATIAL MEMORY FOR EMERGENCY ROBOTS

Team [ANONYMIZED] — Phase 1 Submission

Independent Submission • IEEE RAS × IEEE AESS Tunisia Section TSYP14 Technical Challenge

Abstract—We present *The Living Map*, a distributed spatial-memory system for GPS-denied tunnel search-and-rescue that decouples knowledge acquisition from knowledge retention. A Writer robot explores an unmapped tunnel and externalises its findings into a self-organising chain of RF beacon nodes, each carrying a compact 16-byte record of events, heading, and confidence. An Outside Network Area (ONA) gateway at the tunnel mouth relays aged, GPS-converted events to a remote Command Post, which briefs a second Executor robot before entry. Beacons gossip records toward the exit, so mission knowledge survives Writer failure; a confidence-decay model drives Executor re-verification of aging hazard reports. The Phase 1 simulation framework in Webots establishes an evaluation harness targeting $\geq 80\%$ map coverage, ≤ 0.5 m geodetic localization error, and sub-5 s event-to-dashboard latency across 20 randomized trials.

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Team [ANONYMIZED] — Phase 1 Submission Independent Submission TSYP14 Challenge — IEEE RAS × IEEE AESS — Phase 1 Short Technical Report

ABSTRACT

We present *The Living Map*, a distributed spatial-memory system for GPS-denied tunnel search-and-rescue that decouples knowledge acquisition from knowledge retention. A Writer robot explores an unmapped tunnel and externalises its findings into a self-organising chain of RF beacon nodes, each carrying a compact 16-byte record of events, heading, and confidence. An Outside Network Area (ONA) gateway at the tunnel mouth relays aged, GPS-converted events to a remote Command Post, which briefs a second Executor robot before entry. Beacons gossip records toward the exit, so mission knowledge survives Writer failure; a confidence-decay model drives Executor re-verification of aging hazard reports. Simulation in Webots targets $\geq 90\%$ event detection recall and sub-5 s event-to-dashboard latency over 20 randomised fault trials.

I. INTRODUCTION: CONCEPT, ENVIRONMENT, AND EVENTS

Emergency response in collapsed mines and road tunnels is constrained by three hard realities: no GPS signal, no pre-existing map, and the high probability that the first robot to enter will be damaged or destroyed. Conventional single-robot approaches lose all accumulated knowledge when the platform fails. We address this with *The Living Map*: knowledge is written into the environment itself as a chain of low-cost RF beacons, making it persistent, robot-independent, and accessible to any subsequent responder.

Environment. The challenge environment is a mine/tunnel modelled in Webots [1]. The world contains a fallen beam, a debris field, two static pipes (landmarks), and a gas-leak point. Two event types are defined: **E1 GAS_LEAK** (gas concentration above threshold) and **E2 BLOCKED_PASSAGE** (free width reduced below robot width). A third event slot (e.g. victim/heat source) is reserved in the packet without protocol change. Ground robots only; aerial platforms are not considered.

System roles. The *Writer* robot explores autonomously, detects events, and deposits RF beacons. The *ONA* (Outside Network Area) gateway at the tunnel entrance bridges robot-local RF to an uplink. The

Command Post (CP) fuses events into a live map, plans a mission, and briefs the *Executor* robot before entry.

Compliance by design. All inside-to-outside traffic crosses the ONA exclusively. In simulation, robots and beacons own only Webots Emitter/Receiver devices on three RF channels: ch1 (beacon broadcast), ch2 (Writer-to-beacon write), and ch3 (ONA briefing/reports). The CP is a separate process connected only to the ONA uplink socket; no robot controller opens a network socket, making a direct robot-to-CP link impossible by construction.

II. SYSTEM ARCHITECTURE AND DATA FLOW

II-A. Architecture Overview

Fig. D1 shows the full system architecture. The tunnel contains the Writer, a chain of ≤ 8 beacons, and (during extraction) the Executor. The ONA sits at the tunnel mouth. All information crosses a single interface: the ONA gateway.

SYSTEM TOPOLOGY: THREE-TIER ISOLATION BOUNDARY

GPS-Denied Tunnel (RF-only)

Writer Robot $\rightarrow$ [ch2: write] $\rightarrow$ Beacon Chain $[B_1 \dots B_k]$ (ch1: gossip mesh)

Executor Robot $\leftarrow$ [ch3: briefing] $\leftarrow$ ONA RF Gateway

↓ **Strict Boundary: Encrypted TCP/LTE Uplink (ONA Edge Processor & Store-and-Forward)**

Remote Command Post (Isolated External Process • WGS-84 GPS Frame)

Ingestion Server • Spatiotemporal Event Database • Live Map • Mission Planner

Fig. 1. System tier isolation topology (Full architecture in Fig. D1). Direct robot-to-Command Post communication is physically prohibited.

II-B. End-to-End Data Flow Narrative

Fig. D2 traces the full pipeline. Writer's LiDAR and virtual gas sensor detect an event; the FSM transitions to INVESTIGATE, localises peak concentration or blockage geometry, then transitions to DEPLOY: a 16-byte frame is written via ch2 to a freshly dropped beacon (Fig. D4). The beacon begins broadcasting on ch1 every 2 s (± 500 ms random jitter), piggybacking the three freshest records relayed by deeper beacons (hop-decremented). The ONA RF gateway on ch1 receives these frames, applies CRC-16 validation and de-duplication by (BEACON_ID, TIMESTAMP), runs the local-to-GPS transform (§VI), and enqueues the result in a persistent store-and-forward queue. The uplink delivers the event to the CP ingestion server (emulated by a TCP socket with configurable 2 s latency, 10 % loss). The CP updates its spatiotemporal database, renders each event with confidence-weighted opacity on the live map (Fig. D5), assembles a mission packet, and pushes it back through the ONA to the waiting Executor on ch3 (Fig. D6). The Executor enters, following the inherited beacon chain (Fig. D7), re-verifying AGING events in situ and refreshing timestamps on verified records (Fig. D8).

III. WRITER ROBOT AUTONOMY

Sensing and localisation. Writer carries a 2D LiDAR (360°, 12 m range), IMU gyroscope, and wheel encoders. Heading is gyro-corrected; position is tracked by dead reckoning. Each time Writer passes a previously deposited beacon, it detects the signal-strength peak (known position), re-anchors its pose estimate, and resets the odometry error accumulator. Webots GPS is used only as ground-truth for evaluation metrics; the robot controller never reads it.

Mapping. A 2D log-odds occupancy grid (5 cm cells) is built online from LiDAR scans and the current pose estimate. The grid is used for both path planning and blocked-passage detection (§IV).

Exploration (frontier-based). At each planning tick, the nearest reachable frontier cell (free cell adjacent to unknown) is selected; A* plans a path; a local reactive layer handles transient obstacles. In straight corridors this reduces to advancing deeper; at dead ends it falls back to wall-following. Frontier selection halts when RETURN is triggered.

Decision-making (FSM). The primary FSM (Fig. D3) has four states:

State	Entry condition	Key action
EXPLORE	Default / post-DEPLOY	Frontier-based navigation
INVESTIGATE	Event signal above threshold	Gradient climb, geometry check
DEPLOY	Event confirmed	Drop beacon, write record, ch2 write
RETURN	Budget ≤ 2 , battery $< 25\%$, or no frontier	Navigate to ONA

IV. EVENT DETECTION AND BEACON DEPOSITION

IV-A. Event Type Table

Event	Detection Method	Sensor Used	Beacon Payload Snippet
E1 GAS_LEAK	5-sample moving average of virtual concentration above trace threshold; gradient climb to peak	Virtual gas channel (supervisor-computed, § below)	TYPE=0x1, CONFIDENCE=severity class (0–3), LOCAL_X/Y = peak position
E2 BLOCKED_PASSAGE	Occupied cells persisting for N consecutive scans in previously free space; free width $< 1.2 \times$ robot width	2D LiDAR + occupancy grid	TYPE=0x2, CONFIDENCE=0xFF (binary event), LOCAL_X/Y = blockage centroid
E3 (reserved)	e.g. thermal gradient or victim IR	Plug-in sensor	TYPE=0x3, same frame, no protocol change

E1 — Gas leak emulation. Webots provides no native gas sensor. The supervisor computes concentration $c = c_0 \cdot \exp(-d/L) + \eta$ (Gaussian noise η) from the distance d to the leak node and feeds it to Writer as a virtual sensor channel. Writer applies a 5-sample moving average; exceeding the trace threshold triggers INVESTIGATE. Writer then samples left/centre/right, climbs the concentration gradient, and

stops at a safe stand-off distance, recording the peak position and severity class (0–3 mapped from concentration quartiles).

E2 — Blocked passage. Occupied cells appearing in the occupancy grid where the grid previously held free space, persisting for $N = 5$ consecutive scans and with measured free width $< 1.2 \times$ robot width, raise the BLOCKED flag. The fallen beam and debris field are the primary test cases.

IV-C. Gradient Ascent Gas Source Localization Model

Gas dispersion in the subterranean tunnel is emulated by the Webots Supervisor via an advection-diffusion approximation with additive Gaussian noise: $c(d) = c_0 \cdot \exp(-\frac{d}{L}) + \mathcal{N}(0, \sigma_n^2)$ where $c_0 = 100$ ppm denotes the source concentration, $L = 8.5$ m is the characteristic diffusion length, and $\sigma_n = 2.5$ ppm represents local atmospheric turbulence.

Upon exceeding the detection threshold (≥ 15 ppm), the Writer executes a tri-directional sampling maneuver (sampling $\pm 30^\circ$ relative to heading). The local spatial gradient $\nabla c = (\partial c / \partial x, \partial c / \partial y)$ is approximated numerically: $\nabla c \approx \left(\frac{c_{\text{center}} - c_{\text{rear}}}{\Delta s}, \frac{c_{\text{left}} - c_{\text{right}}}{w_{\text{robot}}} \right)$. The robot drives along ∇c with decreasing step size until $|\nabla c| < \epsilon_{\text{stop}}$, localizing the epicenter within ± 0.4 m before dropping an E1 GAS_LEAK beacon.

IV-B. Beacon Deposition Policy (Budget = 8 Beacons)

Beacon type	Trigger	Beacons reserved
Event beacon	First detection or severity change > 1 class	Up to 6
Link-keeping beacon	Received signal from previous beacon $< 70\%$ radio range	Flexible
Decision beacon	Junction encountered (route ambiguity)	Flexible
Emergency reserve	Last 2 always held for events	2

Drop mechanism. A servo-driven magazine releases one beacon behind Writer. In simulation, Writer transmits a drop request on ch2; the supervisor instantiates a Beacon PROTO (Emitter/Receiver, own controller, own memory) at the dispenser pose.

V. BEACON MESSAGE AND SIGNAL DESIGN

V-A. 16-Byte Frame Schema

Byte(s)	Field	Size	Type	Purpose
0	TYPE[3:0] + FLAGS[7:4]	1 B	bitfield	Event class (0=breadcrumb, 1=GAS_LEAK, 2=BLOCKED, 3–15=reserved); flags: bit4=verified, bit5=low_battery
1–4	TIMESTAMP	4 B	uint32	Mission seconds since T ₀ (clock synced at briefing)
5–6	BEACON_ID	2 B	uint16	Globally unique beacon identifier; used for de-dup by (ID, T)
7	CONFIDENCE	1 B	uint8	0–255 linear map of confidence 0.0–1.0; aged in-beacon and at ONA
8–9	LOCAL_X	2 B	int16	x-position in tunnel frame, centimetres; range ± 327 m
10–11	LOCAL_Y	2 B	int16	y-position in tunnel frame, centimetres
12	LOCAL_THETA	1 B	int8	Heading, units of 1.4° (256 steps over $\pm 180^\circ$)
13	FRAME_ID (low nibble) + TTL (high nibble)	1 B	uint8	FRAME_ID: 0=local, 1=GPS; TTL: hop count, decremented each relay
14–15	CRC-16	2 B	uint16	CRC-CCITT over bytes 0–13; frame discarded on mismatch
Total		16 B		

V-B. RF Signal Design

Beacons broadcast every 2 s with 0–500 ms random jitter on ch1. Emitter range ≈ 6 m models tunnel attenuation and multipath loss. At 9.6 kbit/s, 16 bytes requires ≈ 13 ms airtime, keeping duty cycle below 3 %. Each beacon stores its own record plus the three freshest records heard from deeper beacons; it re-broadcasts all four records with TTL decremented, propagating information toward the exit even if Writer is lost (gossip relay).

V-C. Confidence and Message Aging

Confidence decays as $c(t) = 2^{-(t - T)/\tau}$, with half-life τ chosen per event type:

Event type	Half-life τ	FRESH ($c \geq 0.6$)	AGING (0.25–0.6)	STALE (< 0.25)
GAS_LEAK	10 min	Re-verify only on significant route deviation	Executor slows and re-senses on approach	Not trusted; must re-verify before action
BLOCKED_PASSAGE	30 min	—	Re-verify (debris may have shifted)	Not trusted
Breadcrumb	2 h	—	—	Erase if $c < 0.05$

The CP dashboard renders each event with opacity proportional to c . Beacons erase records with $c < 0.05$ to free relay capacity.

VI. FRAME TRANSLATION: ROBOT COORDINATES TO GPS

VI-A. Coordinate Convention

Writer and Executor share a single private tunnel frame: origin **O** at the ONA reference point, x along the tunnel axis, y to the left, z up. The ONA holds a geodetic anchor (lat_0, lon_0, h_0) and bearing ψ of the

x-axis (clockwise from true north), established from an averaged GPS fix plus two surveyed marks at the tunnel entrance.

VI-B. Transform Pseudocode

Algorithm 1: Geodetic Frame Forward and Inverse Transformations

Forward Transform: $F_{\text{local}}(x, y, z) \rightarrow \text{WGS-84}(lat, lon, alt)$

Input: Local coordinates x, y, z (m); geodetic datum lat_0, lon_0 (deg), h_0 (m); portal azimuth ψ (rad); WGS-84 semi-major axis $R = 6,378,137.0$ m

1: **Local ENU Rotation:**

$$E \leftarrow x \cdot \sin(\psi) - y \cdot \cos(\psi)$$

$$N \leftarrow x \cdot \cos(\psi) + y \cdot \sin(\psi)$$

2: **Spherical Geodetic Projection:**

$$lat \leftarrow lat_0 + (N / R) \times (180 / \pi)$$

$$lon \leftarrow lon_0 + [E / (R \cdot \cos(lat_0 \times \pi / 180))] \times (180 / \pi)$$

$$alt \leftarrow h_0 + z$$

3: **return** (lat, lon, alt)

Inverse Transform: $\text{WGS-84}(lat, lon, alt) \rightarrow F_{\text{local}}(x, y, z)$

Input: Command Post waypoint (lat, lon, alt)

$$1: \Delta N \leftarrow (lat - lat_0) \times (\pi / 180) \times R$$

$$2: \Delta E \leftarrow (lon - lon_0) \times (\pi / 180) \times R \cdot \cos(lat_0 \times \pi / 180)$$

$$3: x \leftarrow \Delta N \cdot \cos(\psi) + \Delta E \cdot \sin(\psi)$$

$$4: y \leftarrow -\Delta N \cdot \sin(\psi) + \Delta E \cdot \cos(\psi)$$

$$5: z \leftarrow alt - h_0$$

6: **return** (x, y, z)

Worked example. Illustrative anchor: $lat_0 = 36.8065^\circ\text{N}$, $lon_0 = 10.1815^\circ\text{E}$, $\psi = 30^\circ$. Gas beacon at (x, y) = (42.0 m, -1.5 m): $E = 22.30$ m, $N = 35.62$ m $\rightarrow 36.80682^\circ\text{N}$, 10.18175°E .

Uncertainty propagation. Dead reckoning drifts with path length s . The ONA attaches uncertainty radius $\sigma(s) = 0.30 \text{ m} + 0.02 \cdot s$ (m) to each translated position. The CP dashboard renders this as a circle around each event marker.

VII. OUTSIDE NETWORK AREA (ONA)

VII-A. ONA Topology

ONA GATEWAY BOUNDARY PIPELINE

Inbound Mesh: Beacon Chain $\rightarrow$ ch1 (LoRa Broadcast) $\rightarrow$ **ONA RF Gateway**

Edge Processing: CRC-16 Check $\rightarrow$ Deduplication $\rightarrow$ Dynamic Aging $\rightarrow$ Geodetic Projection

Outbound Uplink: Store-and-Forward SQLite Queue $\rightarrow$ TCP/LTE Uplink $\rightarrow$ **Command Post**

Briefing Dispatch: Command Post Mission Packet $\rightarrow$ ONA $\rightarrow$ ch3 $\rightarrow$ **Executor Robot**

Fig. 2. Outside Network Area dual-interface pipeline topology (Full schematic in Fig. D7).

The ONA has three logical layers:

1. **Tunnel-side RF gateway** — listens on ch1; writes briefing/reports to Executor on ch3. No IP connectivity; RF only.
2. **Edge processor** — frame parsing, CRC-16 check, deduplication by (BEACON_ID, TIMESTAMP), confidence aging, local-to-GPS translation (§VI), and a persistent store-and-forward queue with ACK and exponential-backoff retry.
3. **Uplink** — In simulation: TCP socket with configurable latency (2 s) and loss (10 %). On hardware: LTE or satellite messaging. All frames re-sent until explicitly acknowledged by CP.

VII-B. ONA Processing Pipeline

Incoming frame $\rightarrow$ CRC check $\rightarrow$ duplicate filter $\rightarrow$ age confidence $\rightarrow$ translate coordinates $\rightarrow$ enqueue $\rightarrow$ uplink with retry $\rightarrow$ ACK. If ACK not received within timeout, frame re-queued with incremented retry count. Dashboard data-age indicator alerts operators when queue is stalled.

VIII. COMMAND POST AND LIVE MAP

The Command Post (Fig. D5) is a separate process that connects exclusively to the ONA uplink socket. It maintains:

- **Event database** — spatiotemporal records keyed by (BEACON_ID, TIMESTAMP), updated on re-verification.
- **Live map** — events rendered in GPS coordinates; marker opacity $\propto$ confidence c ; uncertainty circles $\propto \sigma(s)$; STALE events shown with a strikethrough badge.
- **Mission planner** — operator assigns target event, CP computes an ordered beacon route, identifies no-go segments (FRESH BLOCKED), and packages hazards with confidences.
- **Briefing builder** — assembles a mission packet: {target event, beacon route $[B_1, \dots, B_k]$, hazard list with confidences, no-go segments, mission clock T_0 } and queues it for ONA delivery on ch3.

The CP never communicates directly with any robot; all traffic is mediated by the ONA, enforcing the compliance boundary by design.

IX. EXECUTOR ROBOT

Briefing at ONA entry (ch3). Before entering, Executor receives: mission target, ordered beacon route, hazard list with confidence flags, and T_0 for clock synchronisation.

Navigation FSM.

State	Condition	Action
FOLLOW_CHAIN	Default	Drive toward beacon k using inherited heading/distance; confirm by signal-strength peak; read record
VERIFY	Record is AGING or STALE, or approaching gas zone	Slow down; re-sense; update confidence
EXECUTE	Within 1 m of target event	Measure; write refreshed record with new TIMESTAMP
REPORT	After EXECUTE or abort	Relay updated record via beacon chain / ONA
REROUTE	Segment BLOCKED	Mark BLOCKED; try alternate branch or abort and report

Executor builds its own occupancy grid for local avoidance throughout. The living memory is maintained: Executor refreshes timestamps on verified events, clears records for vanished hazards, and adds newly discovered events. Each updated beacon is re-broadcast, propagating corrections toward the exit.

X. SIMULATION, DEMONSTRATION, AND EVALUATION

Implementation stack. Webots simulation with Python controllers. Repository layout:

Module Path	Functional Description
worlds/living_map_tunne1.wbt	Webots mine world (beam, debris, pipes, gas source)
controllers/writer_robot/	Autonomous frontier exploration, LiDAR SLAM, drop FSM
controllers/executor_robot/	Beacon-guided tracking, verification, living memory update
controllers/outside_network/	Ch1 listener, CRC-16 check, coordinate transform, uplink
src/command_post/	Ingestion server, SQLite event DB, live WGS-84 map
docs/architecture/	Diagrams D1–D8 (Draw.io, Mermaid, SVG, PNG)
tests/	Fault injection suite (Writer failure, gap, collision)

Component status.

Component	Status
Tunnel world (beam, debris, 2 pipes, gas point)	Done — see demo video
Writer: LiDAR mapping, frontier exploration, FSM	Implementation in Progress (Phase 2)
Beacon codec, aging, relay (Emitter/Receiver)	Implementation in Progress (Phase 2)
ONA gateway, frame translation, uplink emulation	Implementation in Progress (Phase 2)
Executor chain-following, CP dashboard	Implementation in Progress (Phase 2)

Evaluation metrics.

Performance Metric	Evaluation Criterion	Design Target	Validation Protocol
Map Coverage	Traversable area accurately mapped in occupancy grid	$\geq 80\%$	2D log-odds grid compared to ground-truth Webots tunnel layout
Grid Alignment (IoU)	Spatial intersection-over-union against tunnel floorplan	≥ 0.70	Bounding polygon overlap computed by Webots Supervisor
Beacon Geodetic Accuracy	Disparity between translated GPS coordinate and ground truth	≤ 0.50 m RMS	Geodetic forward transform verified at surveyed entrance datum
Event-to-Dashboard Latency	Elapsed time from hazard threshold detection to CP live map display	≤ 5.0 s	Multi-hop gossip relay (≤ 3.5 s) + ONA uplink buffer (≤ 1.5 s)
Mission Success Reliability	Executor successfully reaches target and refreshes memory	$\geq 90\%$	Evaluated over $N = 20$ randomized fault-injection trials
Autonomous Transit Time	Transit wall-clock from portal entry to target execution	Target ≤ 60 s	Monitored by Supervisor clock during beacon-guided navigation

XI. FAILURE MODES AND MITIGATIONS

#	Failure	Impact	Detection Signal	Mitigation	Residual Risk
1	Writer fails or loses power	No new exploration	ONA sees last beacon heartbeat age $> 2 \times$ broadcast period	Events live in beacons; gossip relays to ONA; Executor briefed from last state	Unexplored tunnel area
2	Beacon destroyed or battery flat	Chain gap; Executor may miss route	Signal dropout between adjacent beacons	Executor dead-reckons on stored heading/distance; scans for next beacon; falls back to wall-following	Route uncertainty in gap
3	Odometry drift	GPS translation error	$\sigma(s)$ grows beyond threshold; ONA flags high-uncertainty events	Pose re-anchored at each passed beacon; $\sigma(s)$ propagated and shown on map	Residual drift between beacons
4	RF collision or corruption	Lost frame	CRC-16 failure; de-dup detects missing sequence	Random jitter reduces collision probability; repeated broadcasts; retry queue	Very rare undetected bit error (CRC miss-rate $< 10^{-5}$)
5	Uplink / satellite outage	CP map goes stale	ACK timeout; CP data-age indicator	Store-and-forward queue with exponential-backoff retry; Executor operates from last briefing	Operator awareness gap
6	New collapse after Writer passed	Executor path blocked	Executor LiDAR detects occupied cells in planned free space	Executor marks BLOCKED, reroutes or aborts; writes new beacon; CP updated	Mission abort if no alternate path
7	Stale gas hazard	Wrong no-go zone or missed hazard	$c < 0.60$ (AGING flag set)	Executor slows and re-senses on approach; refreshes record	False all-clear if gas dissipated and re-accumulated
8	LiDAR hardware dropout	No mapping or obstacle avoidance	Scan timeout > 100 ms	Switch to wall-following with range sensors at low speed; drop final beacon and return	Blind navigation until exit
9	Clock offset between robots	Wrong confidence ages	Missions clock T_0 distributed at briefing	All ages computed in mission time relative to T_0	Sub-second jitter from serial delivery

XII. IMPLEMENTATION TIMELINE

Period	Milestone
06–19 Oct 2026	Writer autonomy: occupancy grid, frontier exploration, E1/E2 detection in Webots
20 Oct – 02 Nov 2026	Beacon dispenser, 16-byte codec, aging and relay; ONA gateway and frame translation
03–16 Nov 2026	Executor chain-following; CP live map; failure-case tests, $N = 20$ metric runs
17–23 Nov 2026	Hardware prototype: Writer chassis + LiDAR/IMU/gas sensor, ≥ 4 LoRa beacons, ONA gateway, Executor
24–30 Nov 2026	Integration, user manual, pitch deck, repository clean-up
01 Dec 2026	Final submission deadline

Innovation statement. Beacons constitute a distributed, persistent, self-describing spatial memory: each carries a route ('where to go next') plus its finding, allowing Executor to navigate without a shared map and information to survive Writer loss. Confidence-based aging makes the map *live*, quantitatively driving when re-verification is required. Hardware plan: Raspberry Pi / ESP32 controllers, 2D LiDAR, IMU, MQ-series gas sensor, servo magazine, 433 MHz LoRa beacons (Webots Emitter/Receiver parameters matched to LoRa link budget), LTE/satellite gateway at ONA.

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End of Phase 1 Technical Report — Team [ANONYMIZED]